a) This question is on Message-Oriented Middleware (MOM):

i) Describe the THREE key parts of a MOM message. (8

ii) Provide TWO forms of ***Messaging Domains*** in MOM. [6

[2

Q1a)	Do not write colum
i) (Easy) THREE parts of a MOM message are: **-Header** [1 mark] - A header consists of information about the message destination, message expiration time, message unique identification, message delivery mode (persistent or non-persistent) and message time stamp. [1 mark] **-Properties** [1 mark] - Each property item is a tuple consisting of property name and property value pair which are used to extend the functionality of the header [1 mark] **-Body** [1 mark] - The body contents the message and how it will be sent and received. [1 mark]	

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The TWO types of Messaging Domains are: -Point-to-point (PTP) [1 mark] -Publish/Subscribe (Pub/Sub) [1 mark]		
		8 marks

b) Apache ActiveMQ forms the software framework for facilitating the use of Java Message Service (JMS) concepts.

(10 marks)

i) In your own words, explain the role of the Apache ActiveMQ in the JMS programming model.

[2 marks]

ii) What are the TWO important objects in JMS that are maintained administratively? Explain your answer accordingly.

iii) What is the main difference between the ***synchronous*** and ***asynchronous*** type during the consumption of the messages in JMS? **[6 marks]**

[2 marks]

Q1b)	Do not write in this column
i) (Easy)	
-Message broker/JMS provider [1 mark] that facilitates the from more than one client or server [1 mark].	
ii) (Medium)	

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client or server [1 mark].

ii) (Medium)

The TWO Administered Objects in JMS and their descriptions are:

Connection Factory (CF): A connection factory is the object a client uses to create a connection [1 mark] with a provider [1 mark] which encapsulates connection configuration parameters [1 mark]

Destination (D): A destination is the object a client [1 mark] uses to specify the target of messages it produces [1 mark] and the source of message it consumes [1 mark]

iii) (Medium)

Synchronous
—A message consumer calls a `receive()` method to explicitly fetch a message from a destination. [1 mark]

Asynchronous
—A message listener is registered with a message consumer. When messages arrive for the message consumer, they are delivered by calling the listener's `onMessage()` method. [1 mark]

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A message listener is registered with a message consumer. When messages arrive for the message consumer, they are delivered by calling the listener's `onMessage()` method. [1 mark]

	10 marks

c) The servlet class `QueryServlet` with servlet name `ServletCounter` in Figure 1 is implemented in the Deployment Descriptor (DD) and the servlet class is in the directory `com/query` that extends `HttpServlet` class. The URL pattern is `/counter`. Write servlet declaration in your DD for this servlet showing the complete package name of the class including the servlet-mapping and the URL in the DD.

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```
        public void ModelDatabase() {
        System.out.print("<Query response result>");
        }
}
```

Figure 1.

(7 marks)

Q1c) (Hard)	Do not write in this column
<servlet> [0.5 mark]	
<servlet-name>ServletCounter</servlet-name> [1 mark]	
<servlet-class>com.query.QueryServlet</servlet-class> [2 marks]	
[NOTE: <servlet-class> ModuleServlet</servlet-class> is acceptable [1 mark]]	
</servlet> [0.5 mark]	
<servlet-mapping> [0.5 mark]	
<servlet-name>ServletCounter</servlet-name> [1 mark]	
<url-pattern>/counter</url-pattern> [1 mark]	
</servlet-mapping> [0.5 mark]	
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Question 2

a) Examine the Tomcat directory structure in Figure 2 below carefully.

(8 marks)

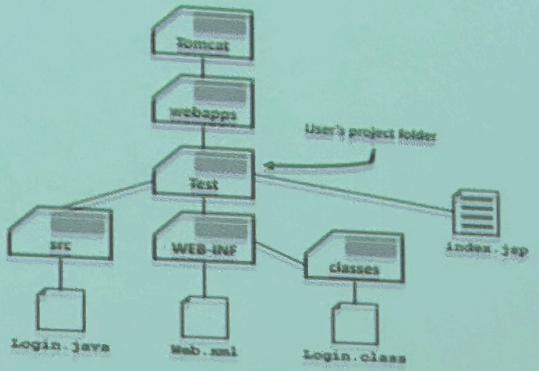


Figure 2.

i) Define the term given to the file "**web.xml**" and explain why it is located under the **WEB-INF** directory.

[2 marks]

ii) In your own words, explain why the "**web.xml**" is needed in the Tomcat Container.

[2 marks]

iii) Provide the correct URL using HTTP protocol and appropriate port number to access the

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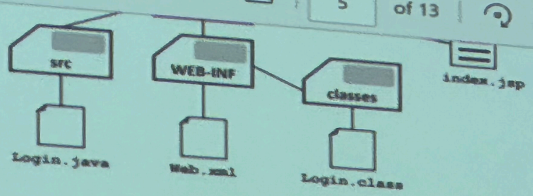


Figure 2.

i) Define the term given to the file "**web.xml**" and explain why it is located under the **WEB-INF** directory.

[2 marks]

ii) In your own words, explain why the "**web.xml**" is needed in the Tomcat Container.

[2 marks]

iii) Provide the correct URL using HTTP protocol and appropriate port number to access the web page "`index.jsp`" if you are running the system installed on your local machine.

[2 marks]

iv) Explain why the file "`index.jsp`" is located directly under the **Project** directory.

[2 marks]

	Do not write in this column
Q2a)	
i) (Easy)	
-This XML file (web.xml) is called the Deployment Descriptor (DD) [1 mark] -It is located directly under the WEB-INF directory so that it will be found	

[2 marks]

Q2a)	Do not write in this column
i) (Easy)	
-This XML file (web.xml) is called the Deployment Descriptor (DD) [1 mark] -It is located directly under the **WEB-INF** directory so that it will be safely secure from unauthorised users (it is not available to the public) [1 mark]	
ii) (Easy)	
[NOTE: Any TWO of the following is 2 marks maximum in total (1 mark each)] -It is used for setting up configurations [1 mark] -It is a configuration file for an artifact that is deployed to some container/engine [1 mark] -It is used for setting up initial parameters [1 mark]	

iii) (Medium)	

iii) (Medium)

The correct URL is: `http://localhost:8080/Project/index.jsp` or `http://127.0.0.1:8080/Project/index.jsp` [2 marks]

iv) (Medium)

-The file `consumer.jsp` is located directly under the **Project** directory because it is available and accessible to the public to access [1 mark].
-It is not restricted by any security measure. [1 mark].

8 marks

b) In your own words, explain the term `Joinpoint` and `Pointcut` terminology for Spring aspect-oriented programming (Spring AOP).

[4 marks]

Do not write in this column

Q2b) (Easy)

Join point

[NOTE: Any ONE of the following is **2 marks** maximum in total (2 marks each)]

- This represents a point in your application where you can plug-in the AOP aspect.

Q2b) (Easy)	this column
Join point	
[NOTE: Any ONE of the following is **2 marks** maximum in total (2 marks each)] - This represents a point in your application where you can plug-in the AOP aspect. [2 marks] - A method invocation. [2 marks]	
Pointcut	
[NOTE: Any ONE of the following is **2 marks** maximum in total (2 marks each)] - This is a set of one or more join points where an advice should be executed. [2 marks] - A set of join points. [2 marks]	
	4 marks

Solution

4 marks

c) This question is about threads.

i) Explain the FIVE important life cycle states of a Java Thread. **(13 marks)**

ii) **Figure 3** shows the Java class `MySubtraction` where the method `getNextValue()` is not thread-safe as it is accessible by two threads. Improve the code by using an intrinsic lock in **Figure 3** to ensure thread-safe and atomic operation. Include the `main()` method to display the output result. **[6 marks]**

[7 marks]

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Q2c)	Do not write in this column
i) (Easy)	
FIVE states of the Java Thread are:	
New: This is when the Thread instance is created using the "`new Thread()`"command. [0.5 mark] The thread is still not alive yet [0.5 mark]	
Runnable: This is when the Thread instance variable invokes the "start()" method using the "`t.start()`" command, where `t` is the thread instance variable [1 mark]. The thread is now alive and has joined the queue of threads about to be executed [1 mark]	
Running: This is when the Thread is now executing [1 mark]	
Blocked/Waiting/Sleeping: This is when the Thread goes out of the execution or running mode when the `wait, interrupt or sleep` commands are executed [1 mark]	
Dead: This is when the Thread completes its task terminates execution calling "`t.destroy()`" command for garbage collection [1 mark]	
ii) (Hard)	
`public class MySubtraction {`	
`    private int value = 10;`	
//[Note: either one of the method (Method 1 or 2) is acceptable [3 marks maximum for each of the method]]	

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[1 mark]

Dead: This is when the Thread completes its task terminates execution calling "t.destroy()" command for garbage collection [1 mark]

ii) (Hard)

```
public class MySubtraction {
    private int value = 10;
//[Note: either one of the method (Method 1 or 2) is acceptable [3 marks maximum for each of the method]]
    // Method 1
    public int getNextValue()[1 mark] {
        synchronized (this) [2 marks] {
            return value--;[1 mark] }}
    // Method 2
    public synchronized [2 marks] int getNextValue()[1 mark]
    {
    return value--; [1 mark] }}

public static void main(String[] args) {[1 mark]
```

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public static void main(String[] args) {[1 mark]	

MyCalculation e =new MyCalculation ();[1 mark]		
System.out.println(e.getNextValue());}[1 mark] *		
}		
* Note: Other forms of output display are acceptable.		
		13 marks

Question marking $\frac{}{8} + \frac{}{4} + \frac{}{13} = \frac{}{25}$

Question 3

a) Describe the sequence of interacting components that enable Middleware Developers to develop highly scalable message-driven applications using Apache Kafka clusters.

[16 marks]

Q3a)	Do not write in this column
The main feature/component in Kafka that enables applications and their infrastructure to scale well is the partitions [1 mark] creating by partitioning the topics. [1 mark]	
Partitioning takes the single topic log [1 mark] and breaks it into multiple logs [1 mark], each of which can live on a separate node [1 mark] in the Kafka cluster. [1 mark]	
This way, the work of storing messages, writing new messages, and processing existing messages [1 mark] can be split among many nodes in the cluster. [1 mark]	
Actions known as events [1 mark] are identified by key/value pairs [1 mark] to store them as logs in Kafka cluster. [1 mark]	
If a message has no key, subsequent messages [1 mark] will be distributed round-robin (even distribution) [1 mark] among all the topic's partitions to scale through the cluster. [1 mark]	
If the message does have a key, then the destination partition [1 mark] will be computed [1 mark] from a hash of the key to scale through the cluster. [1 mark]	
Note: students can also use different words to explain the same concepts.	

Tutors

		16 marks

b) Describe how DDoS security attacks are perpetrated and TWO reasons behind DDoS attacks. **[7 marks]**

	Do not write in this column
Q3b)	
DDoS stands for Distributed Denial of Service [1 mark] and is a type of computer attack coming from multiple sources (distributed sources) [1 mark] intended to "deny" users access to computer resources and services. [1 mark]	
Because the incoming traffic flooding the victim originates from many different sources, this makes it difficult to stop the attack simply by blocking a single source (eg IP address). [1 mark]	
DDoS attackers often target sites hosted on high-profile web servers such as banks or	

credit card payment gateways. [1 mark]	
TWO reasons for DDoS attacks:	
• Revenge, blackmail and activism are some of the reasons/motivations for these attacks, rather than to steal data or money. [1 mark]	
• But of recent, DDoS attacks are used to ask for ransom in money to stop the attack or denial of service. [1 mark]	
NOTE: if students mention other reasons such as: To damage reputation of a big company, government, etc, that is also a valid answer	
	7

c) List TWO constraints of IoT constrained devices.

Q3c
Energy or power constraint [1 mark]
Compute or computational power constraint [1 mark]
NOTE: Other constraints could be: Data storage, lack of IP address, etc

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a) A company intends to use a secure, reusable and runtime environment for developing middleware using Java technology. Describe THREE features in OSGi (Open Service Gateway Initiative) that can meet these requirements.

[6 marks]

Q4a	Do not write in this column
Security (secure means security feature): The security layer that handles authentication, authorisation and protection of bundles [1 mark].	
Components (components are reusable features): A web resource, web service, module or software package that communicates with other interacting components using its interface [1 mark] which is used as reusable modular and cohesive implementation of services. [1 mark]	
Java Virtual Machine (runtime environment): A Java virtual machine (JVM) is a virtual machine that enables a computer to run Java programs including programs written in other languages such as C, C++, C#, etc [1 mark]. The JVM is part of the J2SE/J2EE specification that describes what is required in a JVM implementation [1 mark] to ensure interoperability of Java programs across different platforms [1 mark].	
	6 marks

b) Explain the concepts of Data Integrity and Data Confidentiality. Hence, describe ONE method that can be used to mitigate data Integrity and data confidentiality in computer systems.

across different platforms [1 mark].

6 marks

b) Explain the concepts of Data Integrity and Data Confidentiality. Hence, describe ONE method that can be used to mitigate data Integrity and data confidentiality in computer systems.

[7 marks]

Q4b)	Do not write in this column
Data Integrity is the ability of computer systems to ensure that data at rest and in transit is not changed or manipulated [1 mark] by unauthorised users [1 mark] and that the originality of data must be preserved without anybody tempering with it in an unauthorised way [1 mark].	
Data Confidentiality is the ability of computer systems to ensure that data can only be made available or accessible or seen [1 mark] by only the intended audience or user of the data [1 mark]. Meaning, data should not find its way into the wrong hands [1 mark].	
You can use Cryptography technologies for encryption and decryption of data to ensure data integrity and confidentiality. [1 mark]	
NOTE: Students may describe other methods or tools like using HTTPS, SSH,	

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the data [1 mark]. Meaning, data should not find its way into the wrong hands [1 mark].	
You can use Cryptography technologies for encryption and decryption of data to ensure data integrity and confidentiality. [1 mark]	
NOTE: Students may describe other methods or tools like using HTTPS, SSH,	

physical computer protection, etc		
		7 mar

c) Write a Java program that reads a string directly from the Uniform Resource Locator (URL https://www.qmul.ac.uk/ using the public Java class called URLReader. The program should also print out the string it read from the URL.